

1. (Pascal's Identity) Prove the identity

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

for integers $n \geq 1$ and $1 \leq k \leq n-1$, using a combinatorial argument.

2. Suppose we have n distinct objects and want to distribute them into k labeled boxes, where box i contains exactly n_i objects, with $n_1 + \cdots + n_k = n$. Show that the number of such distributions is

$$\frac{n!}{n_1!n_2!\cdots n_k!}.$$

3. Give a combinatorial proof of the identity

$$\sum_{i=0}^k \binom{m}{i} \binom{n}{k-i} = \binom{m+n}{k}.$$

4. How many solutions in nonnegative integers are there to

$$x_1 + x_2 + x_3 + x_4 = 10?$$

How many solutions are there if we additionally require $x_1, x_2, x_3, x_4 \geq 1$?

5. How many distinct permutations are there of the word

MISSISSIPPI?

6. (Hockey-stick identity) Prove that

$$\sum_{i=k}^n \binom{i}{k} = \binom{n+1}{k+1}.$$